

Full Name: SOLUTIONS

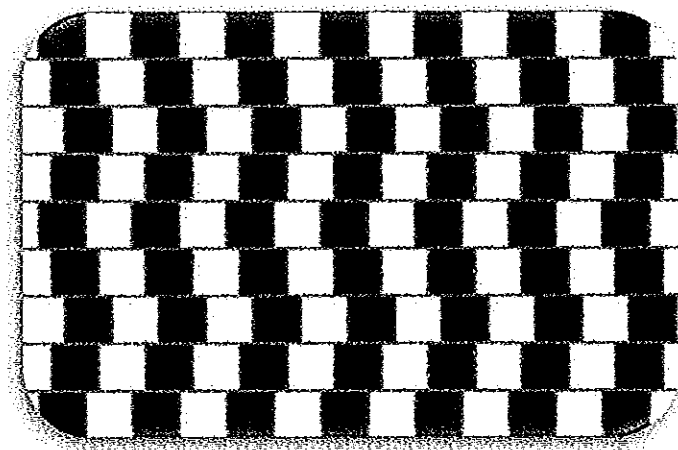


Ellenbrook
CHRISTIAN COLLEGE

**MATHEMATICS
YEAR 08**

Investigation 3 – Angles, Parallel Lines and Transformations

Semester 2 2015

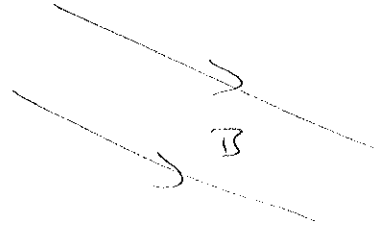
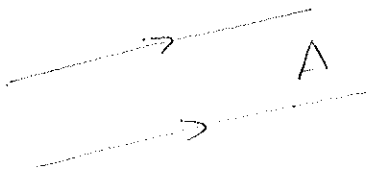


Information for Students

Working time:	70 minutes
Marks available:	56 marks

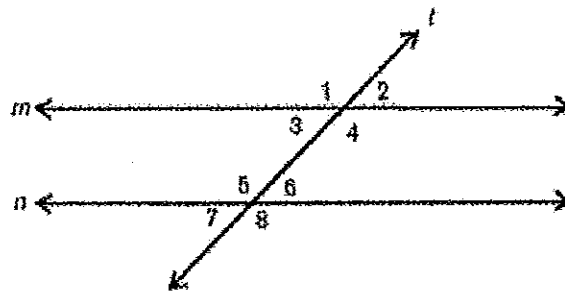
1. (2 marks)

Draw two sets of parallel lines below, labelled set A and set B.



2. (9 marks)

Consider the lines below.



a. What do we know about the gradients (slopes) of lines m and n?

[1]

Same

b. What is the technical name of line t, which cuts lines m and n?

[1]

Transversal

c. What is the term for the relationship between angles 3 and 6?

[1]

Alternate

d. What is the term for the relationship between angles 4 and 6?

[1]

Co interior

e. What is the term for the relationship between angles 2 and 6?

[1]

Corresponding

f. If angle 5 is 120° , how many degrees are in angle 3?

[2]

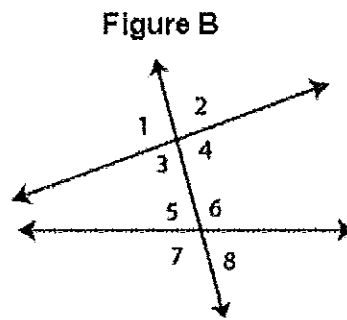
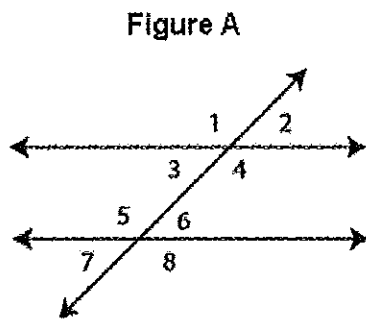
60°

g. If angle 7 is 40° , how many degrees are in angle 4?

[2]

140°

3. (8 marks)



In BOTH figure A and figure B, angle 1 is supplementary to angle 2.

Angle 4 is also supplementary to angle 2.

Supplementary angles always add up to a certain number of degrees, let us call this x°

- a. How many degrees are in x° ? (Consider how many degrees there are in a straight line.) [1]

$$180^\circ$$

- b. In figure B, if angle 6 is 140° , how many degrees are in angle 8? [1]

$$40^\circ$$

In figure A, angles 1 and 4 are **vertically opposite**. So are angles 6 and 7.

- c. If angle 1 is 110° , how many degrees are in angle 2? [2]

What do you then know about how many degrees are in 4?

$$A2 = 110^\circ$$

$$A4 = 110^\circ$$

- d. If angle 6 is 30° , how many degrees is angle 8? How about angle 7? [2]

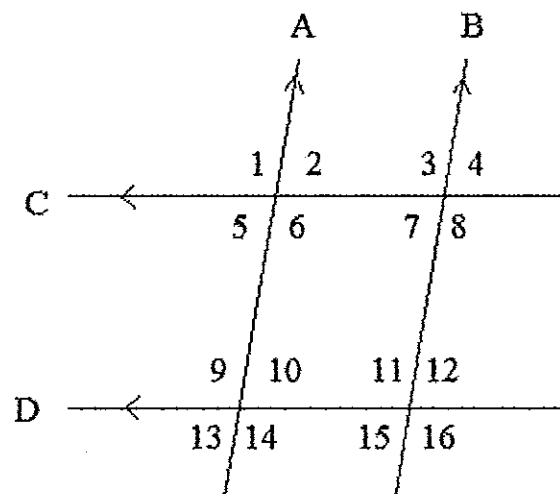
$$A8 = 150^\circ$$

$$A7 = 30^\circ$$

- e. Comparing pairs of angles 1 and 4 and angles 6 and 7 in part a) and b), if we know one angle, what do you think we know about the other vertically opposite one? [2]

Same size

4. (7 marks)



A company wants to design a mosaic using parallelograms. They decide to use the pattern above, but they need to know all the angles involved for the designer. The only piece of information they have is that angle 1 is 100° . Using your knowledge of parallel lines cut by transversals, determine angles 2-16 and write your answers below:

Number	Angle	Number	Angle	Number	Angle
2	80	7	80	12	80
3	100	8	100	13	80
4	80	9	100	14	100
5	100 80	10	80	15	80
6	100	11	100	16	100

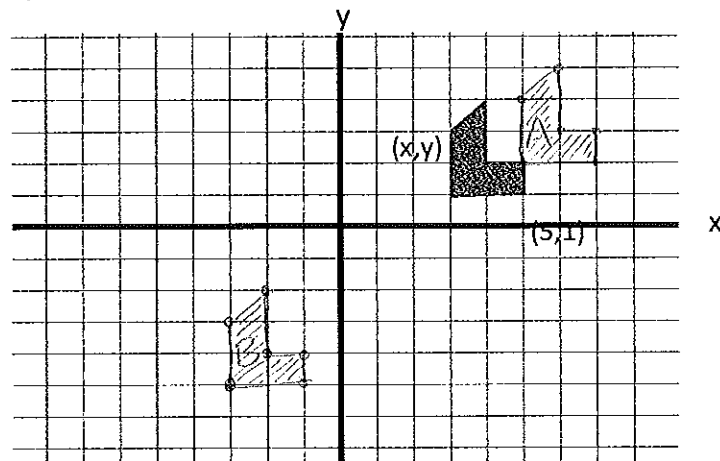
5. (7 marks)

- a. State the (x,y) coordinates of the top-left hand corner of the shape. [1]

$(3,3)$

- b. Translate the shape by $[7,1]$ and label the shape A. [2]

- c. Translate A by $[-8, -7]$ and label the new shape B. [2]



- d. Describe a single translation that could take shape B to shape A. [2]

$[8,7]$

6. (8 marks)

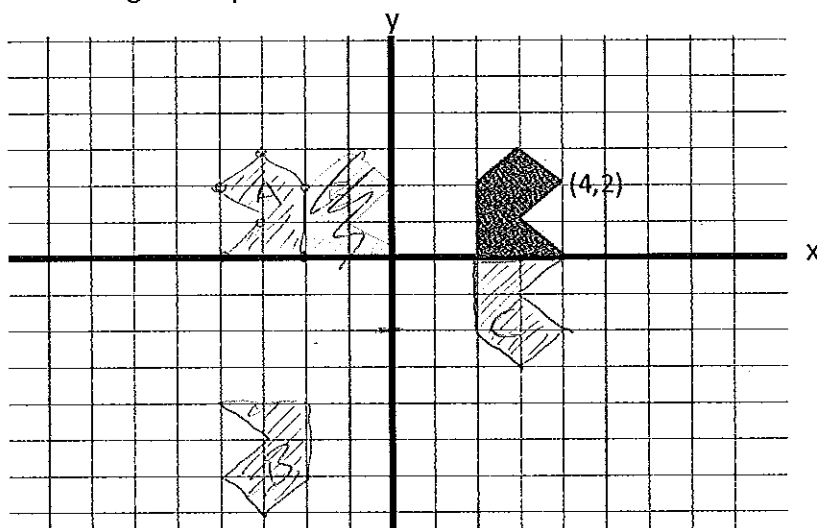
- a. State the (x,y) coordinates of the bottom-left hand corner of the shape. [1]

$(2,0)$

- b. Reflect the shape in the y -axis, and label it A. [2]

- c. Reflect shape A in the line $y = -2$ (draw this line to help) and label it B. [2]

- d. Reflect the original shape in the line x -axis and label it C. [2]



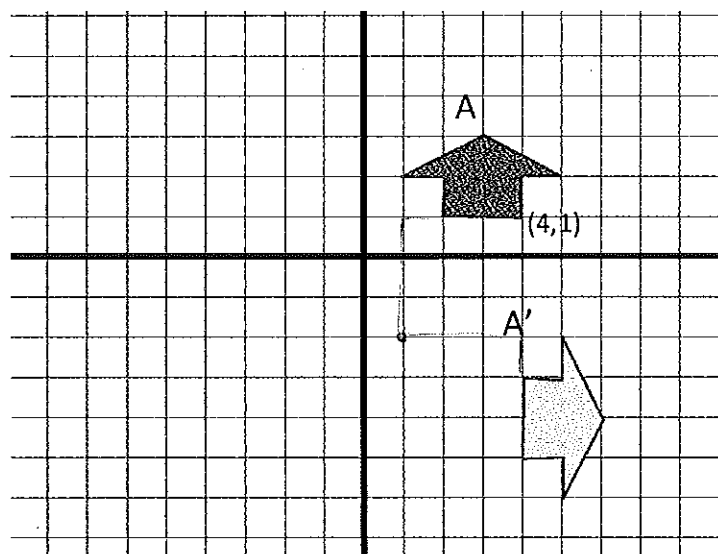
- e. Describe how the x and y coordinates of any point on the outside of the shape (e.g. $(4,2)$) changed when reflecting the original shape in the x -axis [1]

$(4,2)$ become $(4,-2)$

y values change sign (except zeros)

7. (8 marks)

- a. Fully describe the single transformation which took shape A to shape A'. [2]



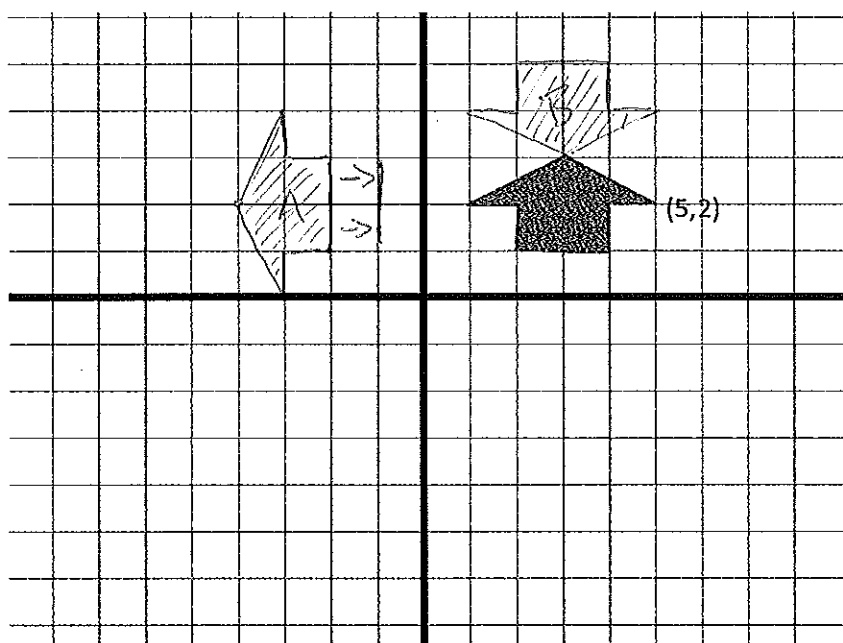
Rotate
90° CW
about $(1,-2)$

- b. Fully describe the single transformation which would return shape A' to A. [2]

Rotate 90° CCW about $(1,-2)$
or
Rotate 270° CW about $(1,-2)$

Rotation

- c. Rotate the shape 270° clockwise about the origin and label it A. [2]
d. Rotate the original shape 180° about the point $(3,3)$ and label it B. [2]



8. (7 marks)

- a. Describe fully the transformation from shape A to the shape B. [1]

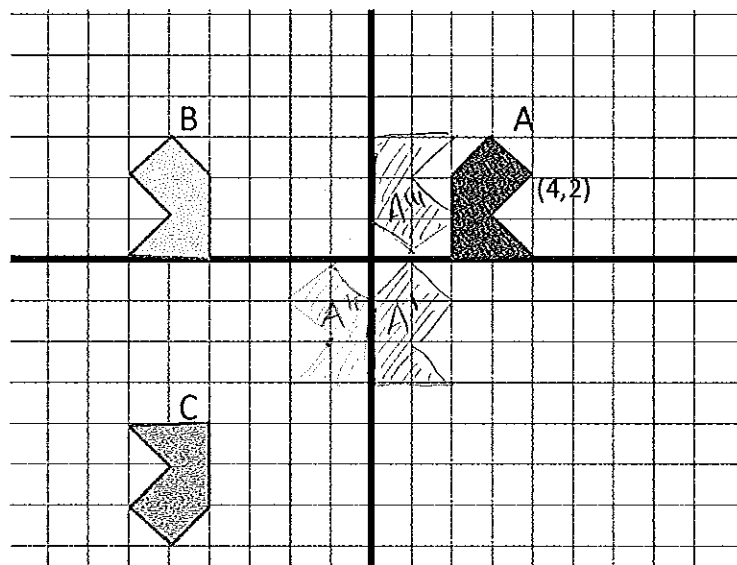
Reflection in $x = -1$

- b. Describe fully the transformation that takes shape B to shape C. [1]

Reflection in $y = -2$

- c. What single transformation could take shape C back to shape A? [2]

Rotation 180° about $(-1, -2)$



- d. Translate shape A by $[-2, -3]$ and label it A' . [1]

- e. Reflect A' in the y-axis and label it A'' . [1]

- f. Rotate A'' 180 degrees about the origin $(0,0)$ and label it A''' . [1]

End of Investigation